

COMMUNICATION

1. f is a polynomial function with a positive leading coefficient of degree n , where n is a positive even integer. Decide whether each of the following statements is true or false. If the statement is false, give an example that illustrates why it is false.

If False provide counter example

a) f is an even function

T or (F)

b) f cannot be an odd function

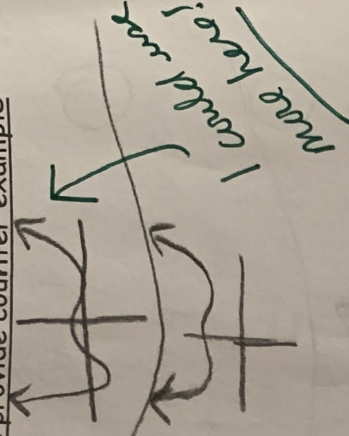
(T) or F

c) f will have at least one zero

T or (F)

d) as $x \rightarrow \infty, y \rightarrow \infty$ and as $x \rightarrow -\infty, y \rightarrow \infty$

(T) or F

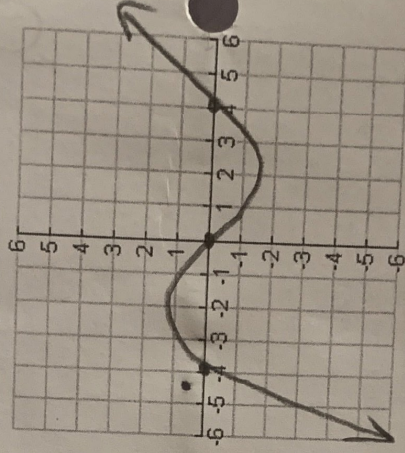


APPLICATION

1. Sketch the graph of the following function $f(x) = 3x^3 - 48x$ on the grid provided.

(2) $f(x) = 3x(x^2 - 16)$

$f(x) = 3x(x-4)(x+4)$



2. A fish dives under water and follows a path modeled by the function

$d(t) = -(t-1)^3 - 3(t-1)^2 - 2$, where d is the depth in meters after t seconds.

- i. Determine the average rate of change from 3 to 4 seconds.

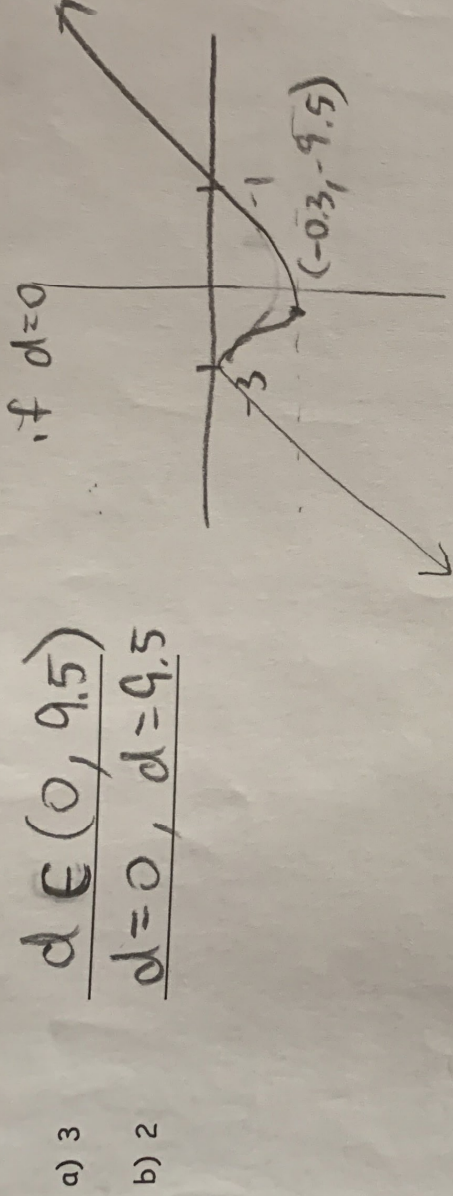
(2)
$$\frac{d(4) - d(3)}{4-3} = \frac{-(4-1)^3 - 3(4-1)^2 - 2 + (3-1)^3 + 3(3-1)^2}{1} = -27 - 27 - 2 + 8 + 12 + 2 = -34 \text{ m/s}$$

- ii.

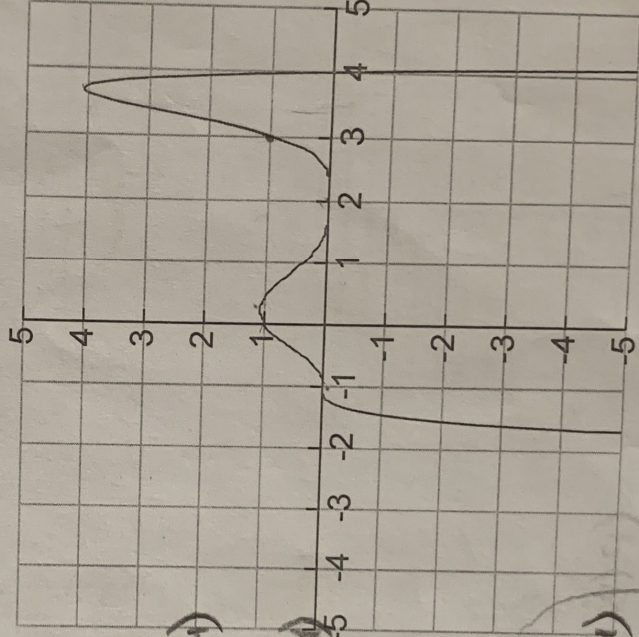
(2)
$$\frac{d(4.001) - d(4)}{4.001 - 4} = \frac{-56.045012001 + 56}{0.001} = -45.0 \text{ m/s}$$

THINKING

1. Consider the polynomial function $f(x) = (x+3)^2(x-1) + d$, where d is a constant. Using interval notation, specify all possible values of d such that there will be the following number of x -intercepts. A local minimum point is approximately $(-0.3, -9.5)$. A quick sketch may help you.



2. Determine the equation of this polynomial function that has a y -int of $(0,1)$.



$$y = a(x+1)^3(x-2)^4(x-4)$$

$$1 = a(0+1)^3(0-2)^4(0-4)$$

$$a = -\frac{1}{64}$$

$$y = -\frac{1}{64}(x+1)^3(x-2)^4(x-4)$$

check with pt $(3,1)$

$$y = -\frac{1}{64}(3+1)^3(3-2)^4(3-4)$$

$$1 = -\frac{1}{64} \cdot 64 \cdot (1) \cdot (-1) \cdot 1 = 1 \checkmark$$

M/C	Written	Total
10/10	24/24	34/34

KNOWLEDGE

- a) For $f(x) = (2-3x)(x+3)^2(x-1)^2$ state the value of the constant finite differences

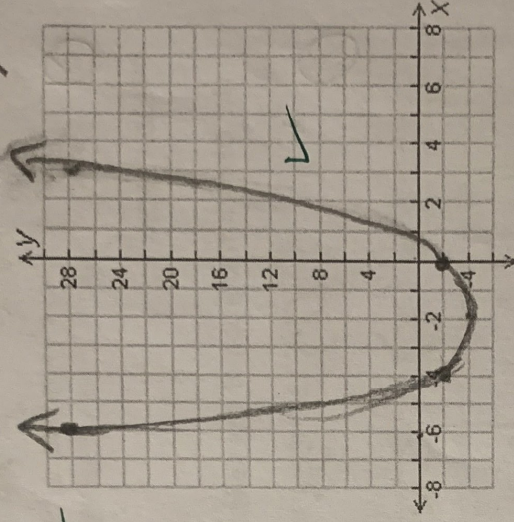
$$a_r = (-3) \cdot (1) \cdot (1) = -3$$

$$CFD = n!a_n = (5!)(-3) = -360$$

2. a) Describe the transformation that must be applied to the graph of the function $f(x) = x^4$,

to obtain the transformed function $y = 2f\left(-\frac{1}{2}x - 1\right) - 4 = 2f\left(-\frac{1}{2}(x+2)\right) - 4$

- Vertical stretch by a factor of 2
- reflection over y-axis
- horizontal stretch by a factor of 2
- left 2 down 4



- b) Sketch the transformed function, showing at least 3 distinct points (must state mapping rule)

$$(x, y) \rightarrow (-2x - 2, 2y - 4)$$

$$(-2, 16) \rightarrow (2, 28)$$

$$(-1, 1) \rightarrow (0, -2)$$

$$(0, 0) \rightarrow (-2, -4)$$

$$(1, 1) \rightarrow (-4, -2)$$

$$(2, 16) \rightarrow (-6, 28)$$

3. The graph of a polynomial function is shown.

- a) Describe the graph in terms of end behaviour.

$$x \rightarrow -\infty, f(x) \rightarrow \infty$$

$$x \rightarrow +\infty, f(x) \rightarrow -\infty$$

- b) Interval(s) where the function is:

i) positive $x \in (-\infty, -2), (-2, -1)$

ii) negative $x \in (-1, 3), (3, \infty)$

- c) Least possible degree? 5

- d) Equation of Function

$$y = a(x+2)^2(x+1)(x-3)^2$$

$$-36 = a(0+2)^2(0+1)(0-3)^2$$

$$a = -1$$

$$y = -(x+2)^2(x+1)(x-3)^2$$

